

Plan: Shell Startup Further Optimization

Baseline: post-zimfw migration — wall time ~1.1s (down from ~9.7s)

Goal: push wall time under 500ms by eliminating the remaining high-cost items

Current hot spots (zimfw zprof)

Rank	Function	Self (ms)	Self %
1	compinit	237	48.7%
2	antigen (residual)	80	16.6%
3	compdef	63	13.0%
4	compdump	55	11.3%
5	compaudit	20	4.2%
6	_zsh_direnv_load	16	3.4%
7	_zsh_highlight_load_highlighters	10	2.1%

compinit + its callees (compdef, compdump, compaudit) account for ~**375ms / 77%** of all tracked time. The **antigen** residual (80ms) signals that zsh-autosuggestions is still loading from the old antigen bundle path.

Fix 1 — Cache compinit (skip rebuild when dump is < 24h old)

Estimated saving: 200–375ms

compinit regenerates ~/.zcompdump unconditionally on every shell. The dump is valid until new completions are installed.

Replace the unconditional call in src/scripts/setup.programs-conf.zsh:

```
# Before
echo 'autoload -Uz compinit && compinit' >> $CLI_CONFIG_PROGRAMS_CONF

# After
cat >> $CLI_CONFIG_PROGRAMS_CONF << 'EOF'
autoload -Uz compinit
if [[ -n ${ZDOTDIR:-$HOME}/.zcompdump(#qN.mh+24) ]]; then
  compinit
else
  compinit -C
fi
EOF
```

-C skips the security audit and dump regeneration when the dump is fresh. This eliminates compaudit, compdump, and most compdef time on every shell except the first one after 24 hours.

Fix 2 — Purge residual antigen path from zsh-autosuggestions

Estimated saving: 80ms

zprof shows antigen still appearing at rank 2. The zsh-autosuggestions bundle is still being sourced from `current/antigen/bundles/zsh-users/zsh-autosuggestions/`. This happens because:

1. `antigen.conf.sh` may still be included in `programs.conf.zsh` (guard on `CLI_CONFIG_MODULES` not yet regenerated), **or**
2. `zimfw's init.zsh` is sourcing the autosuggestions plugin from antigen's cache path.

Steps: 1. Run `cli-config configure && cli-config install` to regenerate `programs.conf.zsh` with `CLI_CONFIG_MODULES=zimfw` — ensures `antigen.conf.sh` is excluded. 2. Verify `zimfw's init.zsh` loads autosuggestions from its own path (`current/zimfw/...`), not `current/antigen/...` 3. Delete `current/antigen/` if it is no longer referenced: `rm -rf $CLI_CONFIG_ROOT/current/antigen`.

Fix 3 — Lazy-load thefuck

Estimated saving: 100–300ms (not yet visible in zprof because thefuck spawns Python *after* zsh init)

`setup.programs-conf.zsh` writes `eval $(thefuck --alias)` into `programs.conf.zsh`. This forks a Python process on every shell.

```
# Replace in setup.programs-conf.zsh:
echo 'eval $(thefuck --alias)' >> $CLI_CONFIG_PROGRAMS_CONF

# With:
cat >> $CLI_CONFIG_PROGRAMS_CONF << 'EOF'
fuck() {
    unfunction fuck
    eval "$(thefuck --alias)"
    fuck "$@"
}
EOF
```

Fix 4 — Lazy-load pyenv init

Estimated saving: 100–200ms (similarly external to zprof — pyenv forks happen after shell functions are set)

`current/conf/pyenv.conf.sh` runs `eval "$(pyenv init -)"` and `eval "$(pyenv virtualenv-init -)"` on every shell.

```
# In pyenv.configure.zsh, replace the eval lines with:
cat >> $CONF << 'EOF'
export PYENV_ROOT="${CLI_CONFIG_ROOT}/current/pyenv"
export PATH="${PYENV_ROOT}/bin:${PYENV_ROOT}/shims:${PATH}"

pyenv() {
    unfunction pyenv
    eval "$(command pyenv init -)"
    eval "$(command pyenv virtualenv-init -)"
    pyenv "$@"
}
EOF
```

Shims cover day-to-day python/pip use. Full init is deferred until `pyenv` is called explicitly.

Fix 5 — Cache oh-my-posh init output

Estimated saving: 30–80ms

current/conf/ohmyposh.conf.sh forks the oh-my-posh binary on every shell to generate the same zsh init script. The output is deterministic for a given theme.

```
# In ohmyposh.configure.zsh, write a loader instead of a direct eval:
cat >> $CONF << 'EOF'
_OMP_CACHE="${CLI_CONFIG_ROOT}/current/ohmyposh/.init.cache.zsh"
_OMP_THEME_MARKER="${CLI_CONFIG_ROOT}/current/ohmyposh/.current_theme"
if [[ ! -f "$_OMP_CACHE" || "$(cat $_OMP_THEME_MARKER 2>/dev/null)" !=
↪ "$CLI_CONFIG_THEME" ]]; then
    "$CLI_CONFIG_ROOT/current/ohmyposh/oh-my-posh" init zsh \
        --config "$CLI_CONFIG_ROOT/current/ohmyposh/themes/$CLI_CONFIG_THEME.omp.json" \
        > "$_OMP_CACHE"
    echo "$CLI_CONFIG_THEME" > "$_OMP_THEME_MARKER"
fi
source "$_OMP_CACHE"
unset _OMP_CACHE _OMP_THEME_MARKER
EOF
```

The cache is invalidated only when CLI_CONFIG_THEME changes.

Fix 6 — Re-enable nvm lazy-load

Estimated saving: 30–100ms

current/conf/nvm.conf.sh adds \$NVM_DIR to PATH but never sources nvm.sh, so the nvm command is unavailable. The lazy-load line in nvm.configure.zsh is commented out.

```
# In nvm.configure.zsh, replace the commented line with:
printf 'lazyload nvm node npm -- "source $NVM_DIR/nvm.sh"\n' >> $CONF
```

Fix 7 — Native CLI_CONFIG_ROOT detection (no forks)

Estimated saving: 2–10ms

Both zshrc profiles resolve CLI_CONFIG_ROOT by forking ls, sed, awk, and xargs:

```
CLI_CONFIG_ROOT=$(ls -la ~/.zshrc | sed "s/^.*\-> //" | awk -F '/' 'NF{NF-=3}1' 'OFS=/' |
↪ xargs)
```

Replace with zsh's native symlink resolution:

```
CLI_CONFIG_ROOT="${${${:-~/.zshrc}:A:h}:h:h}"
```

\${...:A} resolves symlinks in-process; :h strips the last path component.

Fix 8 — Remove .zwc cleanup from default .zshrc

Estimated saving: 5–20ms + fork cost

profiles/default/.zshrc runs on every shell:

```
find ${CLI_CONFIG_ROOT}/current -maxdepth 2 -type f -regex '.*zwc$' | xargs  
→ ${XARGS_OPTIONS} rm
```

This is a one-time install-time cleanup. Move it to `src/defs/install.zsh` and remove it from `profiles/default/.zshrc`.

Priority order

#	Fix	Est. saving	Effort	Dependency
1	Cache <code>compinit</code>	200–375ms	Low	none
2	Purge antigen residual	80ms	Low	regenerate conf
3	Lazy-load <code>thefuck</code>	100–300ms	Low	none
4	Lazy-load <code>pyenv init</code>	100–200ms	Low	none
5	Cache <code>oh-my-posh init</code>	30–80ms	Low	none
6	Re-enable <code>nvm lazy-load</code>	30–100ms	Trivial	none
7	Native <code>CLI_CONFIG_ROOT</code>	2–10ms	Low	none
8	Remove <code>.zwc</code> cleanup	5–20ms	Trivial	none

Fixes 1–2 alone should bring wall time from ~1.1s to ~600–700ms. All eight together should reach < **400ms**.

Measuring

Enable `zprof` in `profiles/mrsauravsahu/.zshrc`: - Uncomment `zmodload zsh/zprof` (line 3) - Uncomment `zprof` (line 132)

Wall time: `time zsh -i -c exit`

Capture before and after each fix. Record results in `results.md`.